Gaia: Smart and Sustainable Forest – INDIA

PRIMARY DOCUMENTATION

Project Name: Gaia: Smart and Sustainable Forest - INDIA

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University: Sri Sai Ram Engineering College, Chennai

INTRODUCTION



Forest Mathigiri, Hosur, Tamil Nadu is a place with huge forest divisions and wildlife connected with the human habitat in various aspects.

“Forest fire breaks out in Cauvery Wildlife Sanctuary, Hosur”, was one of the headlines in the news, on March 17, 2020, at 12:22 a.m. Due to the lack of early detection in the remote forest at Night times, Only 80% of the fire was put out after finding the Location due to the time.

Wildlife Intrusion into Villages - **“Leopard strays into the village, Hosur”** On Jan 12, 2021, 20:13 IST, the Leopard entered the village killing 23 stray dogs, cattle goats, and cows, attacking 5 people and 37 elephants damaged the crops. This occurred due to the lack of detection among the entities.

Wood Theft - Regardless of strong laws, the problem of wood trafficking of sandalwood, teak, and various Timber woods wasn't stopping (nearly 4-5 tractor loads of wood is collected from the forests for cooking, *without permission from the Forest Department*). A proper citation of illegal works is needed. Environmental and Weather Monitoring of forests with an Anti-Theft feature for IoT Nodes is needed

GAIA project has evolved into a fully functional, field-deployed solution designed to support real- time forest monitoring and protection. Initiated under the collaborative efforts of **Sri Sai Ram Engineering College**, **Kenneth Anderson Nature Society (KANS)**, and the **Tamil Nadu Forest Department (Cauvery North Wildlife Sanctuary division) - Government**, this project was aimed at addressing the increasing threats of **forest fires**, **wildlife intrusions**, and **illegal timber trafficking**. **SOURCE**

CITATION LINKS: PROBLEM STATEMENTS , SITE 2, SITE 3, SITE 4

DEPLOYMENT REGION – METHAGIRI, HOSUR -INDIA





Methagiri Forest in Hosur, Tamil Nadu, is part of the ecologically sensitive Cauvery North Wildlife Sanctuary, known for its rich biodiversity and dense vegetation.

The solution that we designed was the deployment of fully connected Lorawan systems in a cluster network topology with the IoT Sensor nodes for parameter monitoring, IoT nodes for camera-based detection and data analysis for making predictive decisions, and finally the centralized Lora-wan Gateway to connect and transmit the data to the web dashboard application for the end user ultimately. The Total NGO adopted area of 504.334 sq. km, where 2-kilometer radius forest area that lies immediately near the villages which have a human habitat of approx. 8000 people living near it, those are the people who get affected by it to the worst extent due to the lack of early monitoring system. Our solution is to connect the Lorawan monitoring systems with the NGO and forest department center for the safety and wellness of the people. Since the Lorawan module has coverage up to a 25km radius range for data transmission, it's perfect to use it for this project.

SOURCE CITATIONS LINKS: IMPACT AREA ANALYSIS, GOOGLE MAPS, GOOGLE EARTH METRICS

Expected Field Deployment Area – Mountain & Dense Forest Terrain





SOURCE IMAGE: KANS NGO SITE

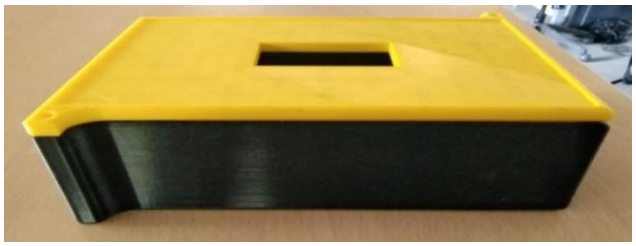
Deployment System Design

Sensor node installation

INITIAL ENCLOSURE DESIGN FOR GROUND NODE,

Top & Side View,

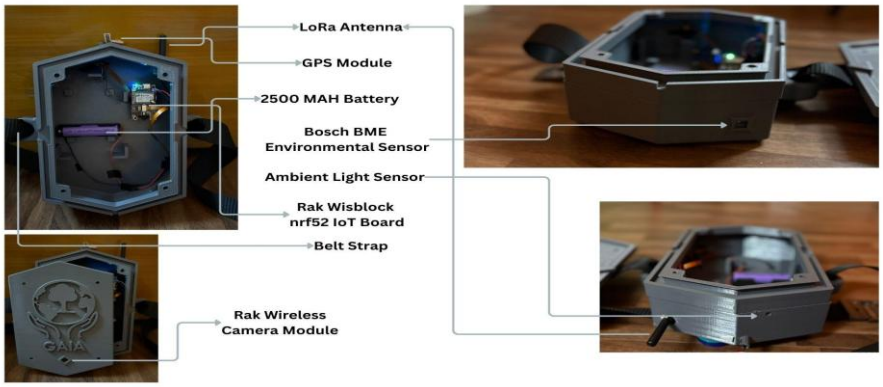




IoT Node after Ground Deployment,

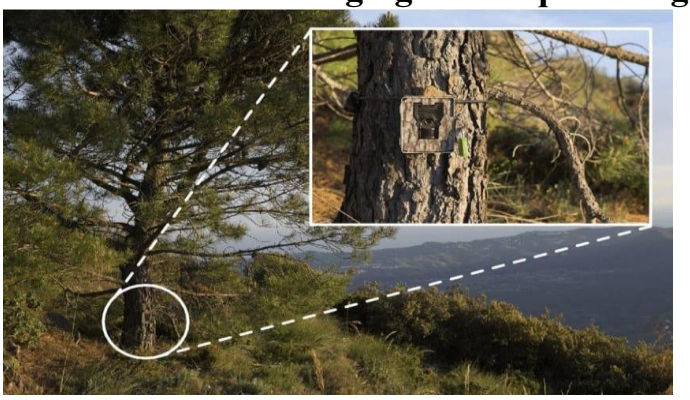


INITIAL ENCLOSURE DESIGN FOR AERIAL NODE,





LoRA Node Camouflaging Technique – Engraving Inside Tree Bark



1) **Natural concealment** – Blends seamlessly with tree bark.
2) **Protection** – Shields from weather and animal interference.
3) **Anti-theft** – Harder to detect and tamper with.
4) **No visual pollution** – Preserves forest aesthetics.
5) **Reduces disturbance** – Minimizes human and wildlife attention.

NOTE: IDEA APPROVAL PENDING DUE TO FOREST RESTRICTIONS

New Revamped Enclosure design –Industrial Grade

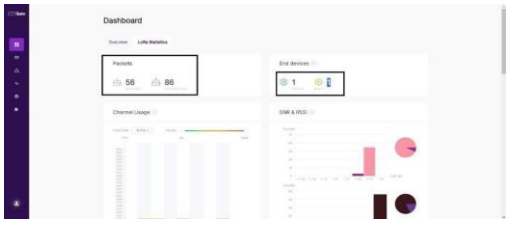
IoT Node after Aerial Deployment,





GATEWAY AND IOT NODE IMAGES





Technical Overview of SAFT LS 14500 12.8V 2400mAh Li-$SOCl_2$ Battery Pack,



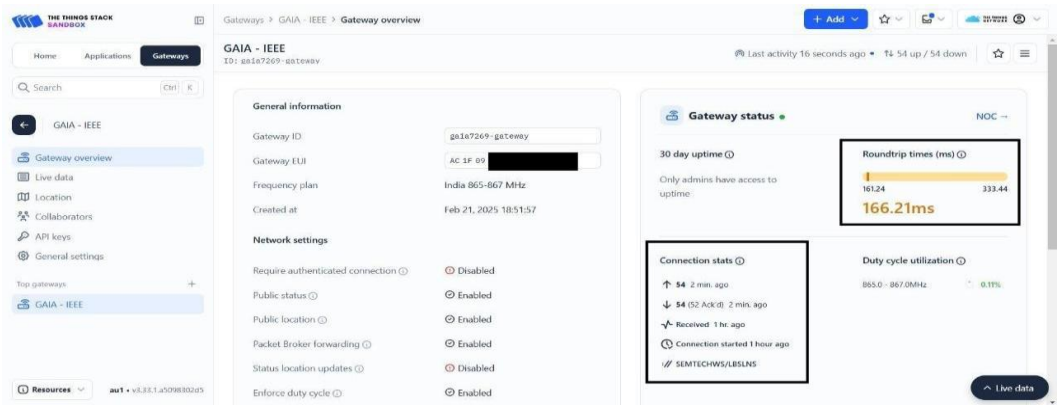
The SAFT LS 14500 is a high-density, long-lasting 3.6V lithium-thionyl chloride (Li-$SOCl_2$) battery with reliable energy over a long term for applications that need it. Constructed in a standard AA-size package, the battery has a typical capacity of 2.6Ah and is renowned for its superior energy density, low self-discharge characteristic, and long shelf life of up to 10 years.

Designed with longevity in mind, the LS 14500 can run within a broad temperature range of -60°C to +85°C and is perfect for difficult and extreme conditions. It provides stable voltage output and has consistent performance, even in low current drain applications.

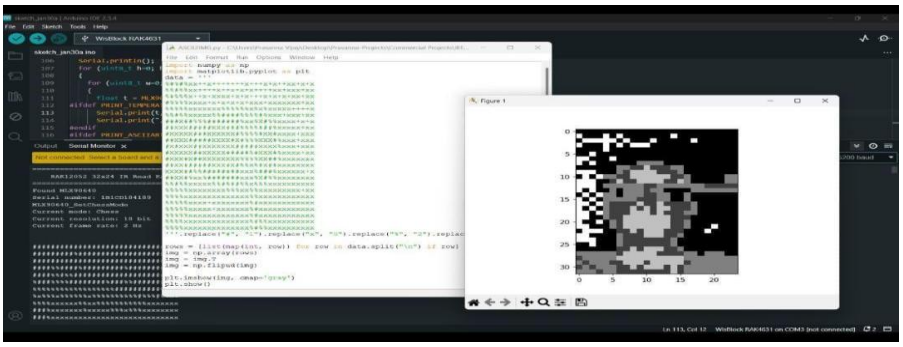
Widely applied in IoT devices, utility metering, industrial sensors, medical devices, and backup systems, the SAFT LS 14500 provides reliability where maintenance and battery replacement are not possible. Its well-documented history of use in critical applications makes it a designer's go-to for long-life, stable, and safe power solutions.

Parameter	Details
Operating Voltage	10 V – 14.6 V
Duty Cycle (Target)	~1–5% (e.g., 1–3 minutes ON per hour)
Runtime (Goal)	**Up to 6 months** (with ultra-low power design)

LoRa Gateway Connection to Things Stack Network Server,



Infrared Matrix Camera and MLflow-Based CV Pipeline Deployment to Azure Cloud

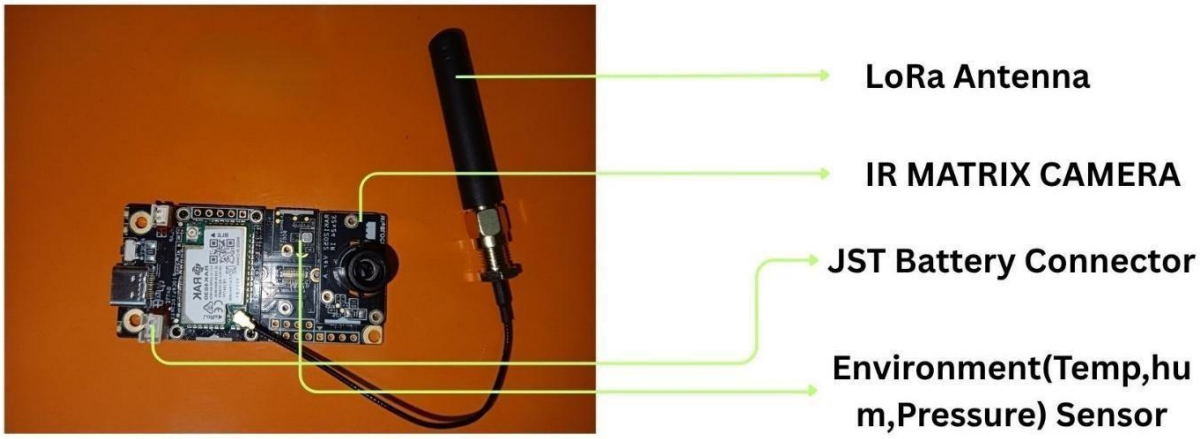


We have been working with RAK Infrared (IR) matrix camera, which functions as a low- power IR sensor that outputs data as ASCII representing pixel intensity points across a 2D array. The raw IR data is processed using NumPy for matrix manipulation and structured into a 2D format. This matrix is then visualized using Matplotlib, converting the numerical data into a grayscale heatmap image.

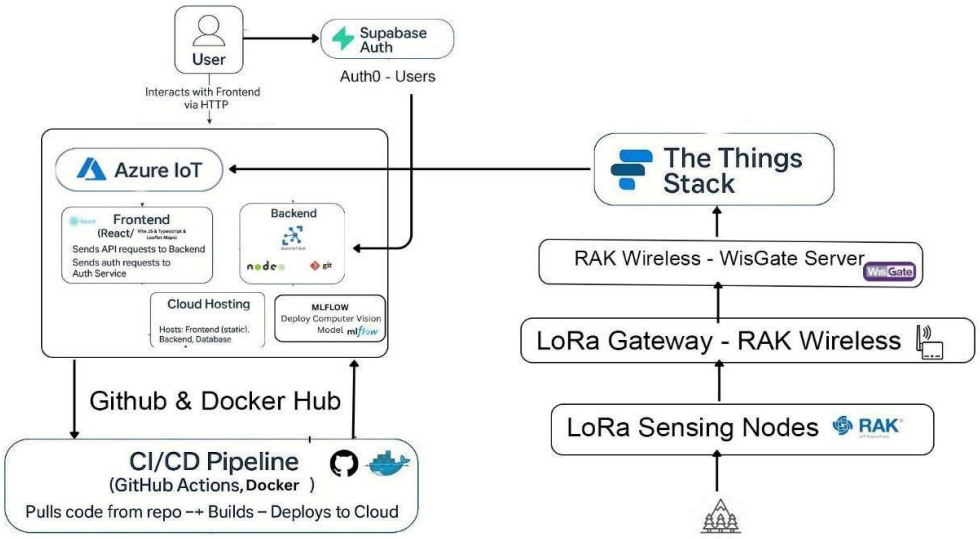
NOTE: The above image is the human Head, (particularly head pose) represnting thermal outlines.

This entire data processing and visualization pipeline is designed to run on the backend server, while the web application dynamically renders the images after fetching preprocessed data.

Then a Computer Vision Model (Using CNN or YOLO) will be deployed in Azure Cloud, we are integrating MLflow to manage and deploy this computer vision model.

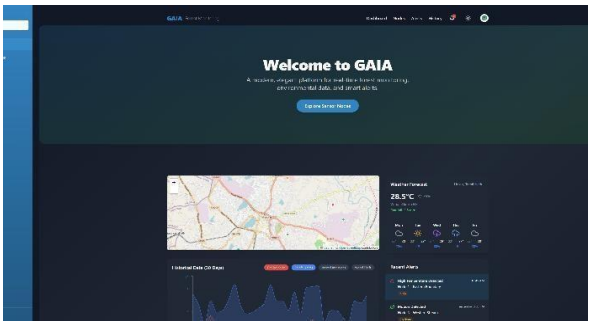


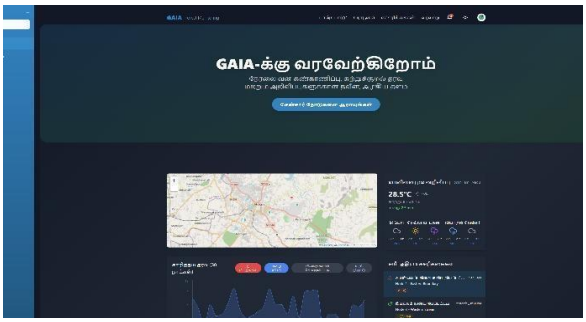
Website Architecture :

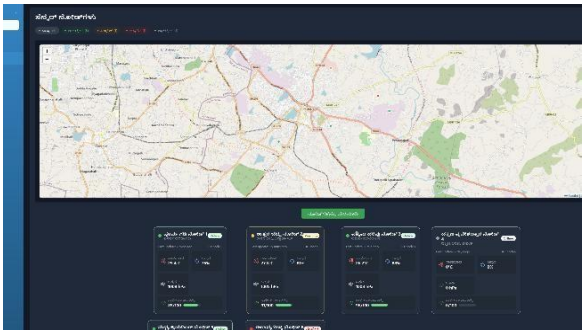


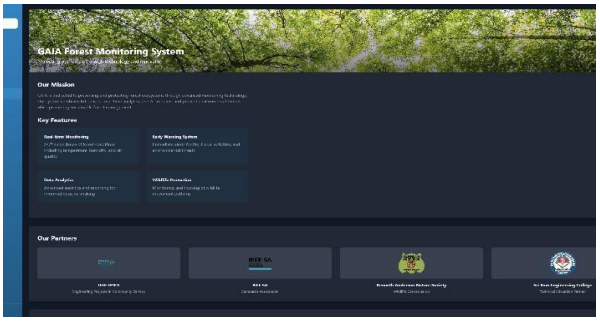
Web dashboard for forest Rangers and NGO

Frontend - English and Native Tamil & Kannada Language version







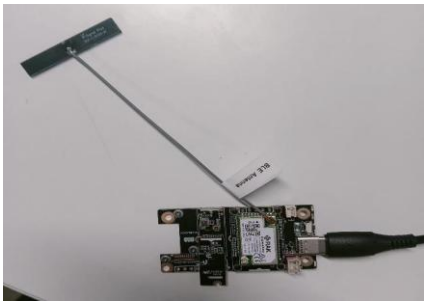


IMAGES OF STUDENT WORKING ON PROJECT GAIA









INTERACTIONS WITH EXTERNAL ENTITIES AND DIGNITARIES

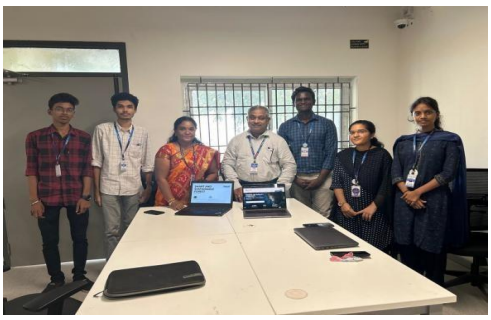
Interaction with Mr. Sampathkumar Veeraraghavan (2023 IEEE-HKN Global President)





A milestone in our outreach was an interaction with **Mr. Sampathkumar Veeraraghavan**, **2023 IEEE-HKN Global President & IEEE Humanitarian Technologies Board (HTB) Past Chair**. He appreciated the humanitarian potential of the GAIA system and advised us to pursue long-term sustainability and replicability. His input directly influenced our efforts toward **compliance with international humanitarian tech standards**.

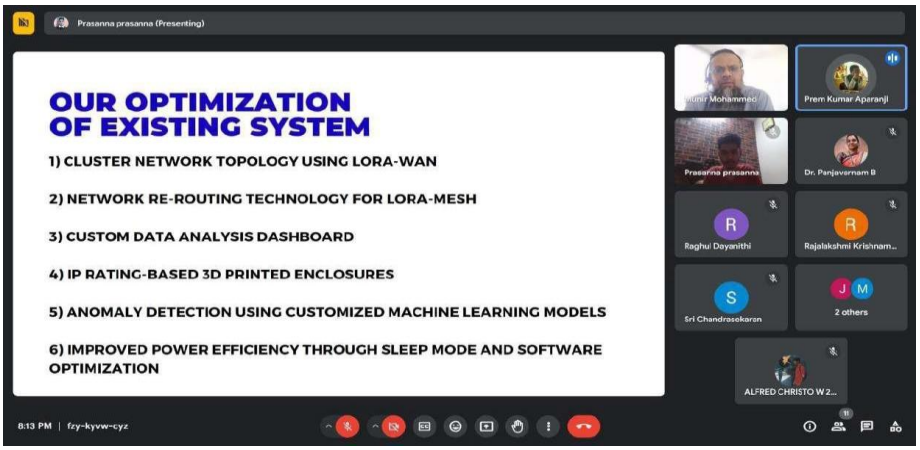
Engagement with Dr. S.K. Uma Maheswaran (UBA Coordinator, 2023, College Chapter)





Our early direction was significantly shaped by an interaction with **Dr. S.K. Uma Maheswaran**, the **Unnat Bharat Abhiyan (UBA – Social Service Scheme by Indian Govt) College Chapter Coordinator (2023)**. He emphasized the importance of community-oriented innovation and encouraged us to integrate feedback mechanisms from local stakeholders. This led us to prioritize ranger and villager usability in the design, especially in the **dashboard UI/UX** and multilingual support.

Interaction with Mr. Munir Mohammed (Senior Manager, IEEE Standards Association)



We also engaged with **Mr. Munir Mohammed**, **Senior Manager – Product Development & Market Engagement** at the **IEEE Standards Association**. His feedback on aligning the system with **IEEE environmental sensing and interoperability standards** prompted us to finalize **LoRaWAN** for low-power communication. He acknowledged the project's potential to influence broader IEEE standardization for rural tech deployments and encouraged inclusion of IEEE standards as protocol in our project.

SOCIAL IMPACT

The GAIA Smart Forest Monitoring System has created a significant positive impact on both the environment and the local communities residing near forest boundaries.

Beneficiaries:
The project directly benefits **over 8,000 people** living in and around the **Anchetty–Mellagiri Hills region**, where human settlements closely interact with forest ecosystems.

Reduction in Human–Wildlife Conflict:
Continuous monitoring and early detection systems help alert authorities about animal movement (such as elephants and leopards), reducing unexpected encounters and preventing loss of life, livestock, and crops.

Early Forest Fire Detection:
The deployed IoT sensor network enables **real-time environmental monitoring**, allowing early identification of fire risks, especially during night-time or in remote regions. This significantly improves response time and minimizes damage.

Prevention of Illegal Activities:
The system supports **monitoring of illegal logging, wood theft, and poaching activities**, enabling authorities to take faster and more targeted action.

Support to Forest Authorities:
Real-time data transmission through LoRaWAN and cloud-based dashboards enhances decision-making capabilities for forest rangers and NGOs, improving operational efficiency.

FUTURE SCOPE

The GAIA project is designed with scalability and sustainability in mind, and several enhancements are planned for future development:

Expansion of Coverage Area:
The system is planned to scale up from the current deployment to cover **up to 100 km of forest area**, significantly increasing the monitoring range.

Increased Sensor Network Density:
Deployment of **20 or more IoT sensor nodes** across the extended region to improve data accuracy, redundancy, and reliability.

Solar-Powered IoT Nodes:
Future nodes will incorporate **solar energy systems**, enabling:

- Longer operational life
- Reduced maintenance and battery replacement
- Sustainable and eco-friendly power solutions

Enhanced AI-Based Detection:
Integration of advanced **AI/ML models** for:

- Improved wildlife detection accuracy
- Behavior prediction
- Fire risk forecasting

Replication Across India:
The GAIA framework can be replicated in other forest regions across India, making it a **scalable model for national-level smart forest monitoring systems**.

THANK YOU